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NVIDIA Jetson Nano Developer Kit

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NVIDIA Jetson Nano Developer Kit

NVIDIA Jetson Nano Developer Kit I

The **NVIDIA Jetson Nano Developer Kit** is a compact and powerful embedded computing platform designed for edge AI and robotics. It enables developers to run multiple neural networks in parallel for tasks such as object detection, segmentation, and speech processing.NVIDIA, 2020b

NVIDIA Jetson Nano Developer Kit II

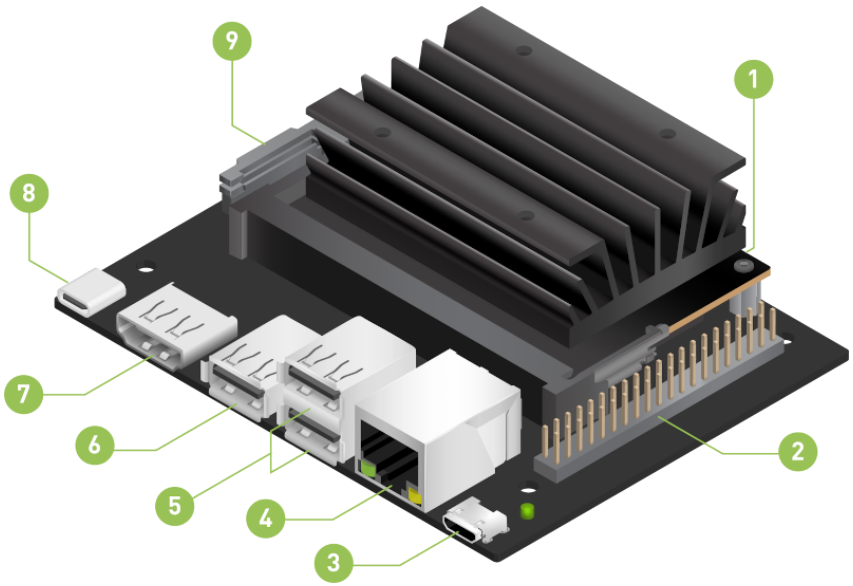


Abbildung: NVIDIA Jetson Nano Developer KitNVIDIA, 2020a
1. microSD card slot for main storage, 2. 40-pin expansion header, 3. Micro-USB port for Device Mode,
4. Gigabit Ethernet port, 5. USB 2.0 ports (x2), 6. USB 3.0 port (x1),
7. HDMI output port, 8. USB-C for 5V power input, 9. MIPI CSI-2 camera connector

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Key Features I

- GPU-accelerated edge computing with low power consumption.
- Full Linux (Ubuntu-based) development environment.
- Support for NVIDIA JetPack SDK and TensorRT.
- Compatible with many open-source AI frameworks such as TensorFlow, PyTorch, and OpenCV.

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Datasheet Reference

Official Reference I

For full specifications and design information, refer to the official NVIDIA datasheet:

- NVIDIA Corporation, *NVIDIA Jetson Nano Developer Kit — Data Sheet*, Version 1.1, 2020.
- Available at: Developer Kit

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Hardware Specifications

Core Hardware I

- **GPU:** 128-core NVIDIA Maxwell architecture.
- **CPU:** Quad-core ARM Cortex-A57 @ 1.43 GHz.
- **Memory:** 4 GB 64-bit LPDDR4 @ 25.6 GB/s.
- **Storage:** microSD card slot for system and user data.
- **Networking:** Gigabit Ethernet (10/100/1000 BASE-T).
- **Display:** HDMI 2.0 / DP 1.2, up to 4K @ 60 Hz.
- **Camera:** MIPI CSI-2 x2 interfaces.
- **Video Encode:** 4K @ 30 fps (H.264/H.265).
- **Video Decode:** 4K @ 60 fps (H.264/H.265).
- **I/O:** 40-pin GPIO header (I^2C, SPI, UART, I^2S, GPIO).
- **Power:** 5 V DC (micro-USB or barrel jack input).

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Core Hardware II

Power Modes

- 5 W power mode for low-power applications.
- 10 W mode for high-performance workloads.

NVIDIA Jetson Nano Developer Kit Carrier Board I

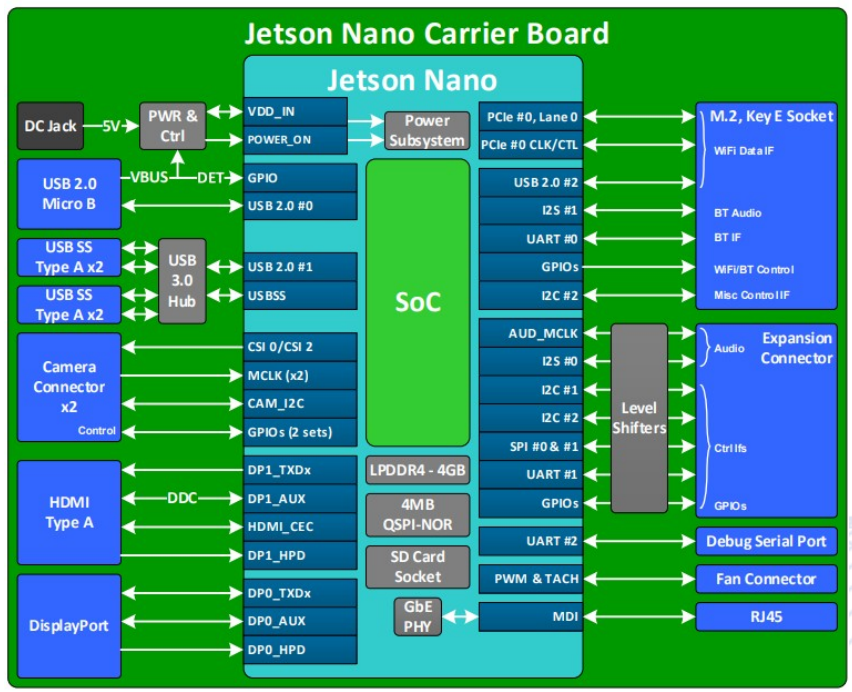


Abbildung: NVIDIA Jetson Nano Developer Kit Carrier BoardNVIDIA, 2020a

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Sensors in NVIDIA Jetson Nano Developer Kit

Sensors in NVIDIA Jetson Nano Developer Kit I

The Jetson Nano supports a variety of sensor interfaces (e.g., I²C, SPI, UART, GPIO) enabling connection of many sensors for e.g. temperature, humidity, motion, proximity. According to Kurniawan (2021) it "can implement IoT solutions ... with the board having GPIO pins and a GPU core". Although specific sensor modules are external, typical sensors used with Jetson Nano include IMUs, LiDARs, ultrasonic distance sensors, environmental sensors, etc. For these sensors you would write code to interface over I²C/SPI or via USB.

Testing codes for all the sensors

Testing codes for all the sensors I

After connecting sensors to Jetson Nano, you would write test code. For example, using Python with libraries such as `smbus2` for I²C, `spidev` for SPI, `serial` for UART, and GPIO via `Jetson.GPIO`. Micro-example:

```
/**
 * @file motion_detection.py
 * @brief Simple script to detect motion using a GPIO pin on
     a NVIDIA Jetson board.
 * @author NVIDIA
 * @date 2025-11-06
 * @details This script initializes BCM pin 17 as an input
    and checks its current state.
 *          It prints "Motion detected" if the pin is high,
    otherwise "No motion".
 */

import Jetson.GPIO as GPIO
import time

```

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Testing codes for all the sensors II

```
13 # Disable warnings that might appear when re-running the
       script
14 GPIO.setwarnings(False)
15
16 # Set the pin numbering mode to BCM (Broadcom SOC channel
       number)
17 GPIO.setmode(GPIO.BCM)
18
19 # Configure pin 17 as an input pin
20 # The internal pull-up/down resistors can be set here if
       needed, e.g., pull_up_down=GPIO.PUD_UP
21 GPIO.setup(17, GPIO.IN)
22
23 # Check the current input status of pin 17
24 if GPIO.input(17):
25     # The pin is currently high (3.3V)
26     print("Motion detected")
27 else:
28     # The pin is currently low (0V)
29     print("No motion")
```

Testing codes for all the sensors III

```
30
31 # It is good practice to clean up the GPIO channels when the
       script finishes.
32 # Uncomment the line below when implementing this in a
      production environment or a loop.
33 # GPIO.cleanup()
```

You would test each sensor: initialise it, read raw values, process/validate them, log results. For IoT sensor-reading with Jetson Nano see e.g. the study on agricultural machines: "Engineering of Information Monitoring System Sensor Reading Data Based on Smart Wireless using NVIDIA Jetson Nano and Arduino Mega. . ."

Compatible Camera for NVIDIA Jetson Nano Developer Kit

Compatible Camera for NVIDIA Jetson Nano Developer Kit I

The Jetson Nano supports MIPI CSI-2 camera modules (2 lanes) and USB cameras. The book by Kurniawan (2021) mentions "the board ... enable[s] stronger IoT solutions ... with integrated sensor and actuators" and includes a chapter on the camera. When selecting a camera ensure driver support under the OS and suitable resolution/frame rate for your application (e.g., 1080p @30 fps or better for real-time vision).

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OS compatible with NVIDIA Jetson Nano Developer Kit

OS compatibility I

The NVIDIA Jetson Nano Developer Kit officially uses the NVIDIA JetPack SDK (which includes Ubuntu Linux, CUDA, cuDNN, TensorRT). According to NVIDIA Jetson Nano Datasheet, you can install Ubuntu 18.04/20.04 based images provided by NVIDIA. Some community builds support other Linux distributions but for full GPU/vision support JetPack is recommended.

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OS installation in NVIDIA Jetson Nano Developer Kit

OS Installation I

Here is a high-level installation process:

- Download the JetPack image from NVIDIA's website.
- Flash the microSD card (or eMMC/module depending variant) with the image.
- Insert into Jetson Nano, connect keyboard/monitor, power it up.
- Follow on-screen initial setup (locale, timezone, user, WiFi/Ethernet).
- Run the SDK Manager or apt-install additional packages (CUDA, cuDNN, TensorRT).

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Setup after OS Installation in NVIDIA Jetson Nano Developer Kit

Setup after OS Installation I

After OS installation you should perform:

- Update and upgrade the system: `sudo apt update && sudo apt upgrade`.
- Enable necessary services (e.g., ssh, VNC).
- Test GPU/CUDA: run `nvidia-smi` or the Jetson Stats tool.
- Install camera drivers and verify camera input (e.g., `gst-launch-1.0` for CSI camera).
- Configure sensor interfaces: enable I^2C, SPI in device tree or via `/boot/firmware/config.txt` if needed.
- Clone/test sample code (e.g., for object detection) — many papers show such usage: see e.g. “Benchmarking Deep Learning Models on NVIDIA Jetson Nano”

Connector cables compatible with NVIDIA Jetson Nano Developer Kit

Compatible Connector Cables I

Typical connector cables you may need:

- Micro-USB or USB-C (depending variant) for power supply.
- 5 V/4 A power supply recommended.
- MIPI CSI-2 ribbon cable (15-pin) for compatible camera modules.
- HDMI cable or DisplayPort cable for monitor output (depending variant).
- I^2C/SPI/UART jumper wires to GPIO header (40-pin) for sensors.
- USB 3.0 cables for external USB cameras or USB sensors.

Make sure cable lengths are reasonable to avoid voltage drop or signal integrity loss.

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NVIDIA Jetson Nano Developer Kit Applications

Applications I

Here are some example applications of the Jetson Nano:

- Real-time pedestrian and priority sign detection (vehicle-embedded).Sarvajcz, Ari und Menyhárt, 2024
- Crack detection on concrete surfaces via transfer learning on edge device. Nguyen u. a., 2024
- Remote insect monitoring for agriculture using YOLOv7. Doan und Phan, 2024
- Sensor-data monitoring with Jetson Nano + Arduino on agricultural spraying machines. Siskandar, 2023
- Benchmarking deep learning models for real-time systems on Jetson Nano.Swaminathan, Silver und Akilan, 2024

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Waveshare Case Assembly on Seedstudio Jetson Nano J1010

Waveshare Case Components I

Jetson Nano Case (C) Assembly

Check the components before assembling.

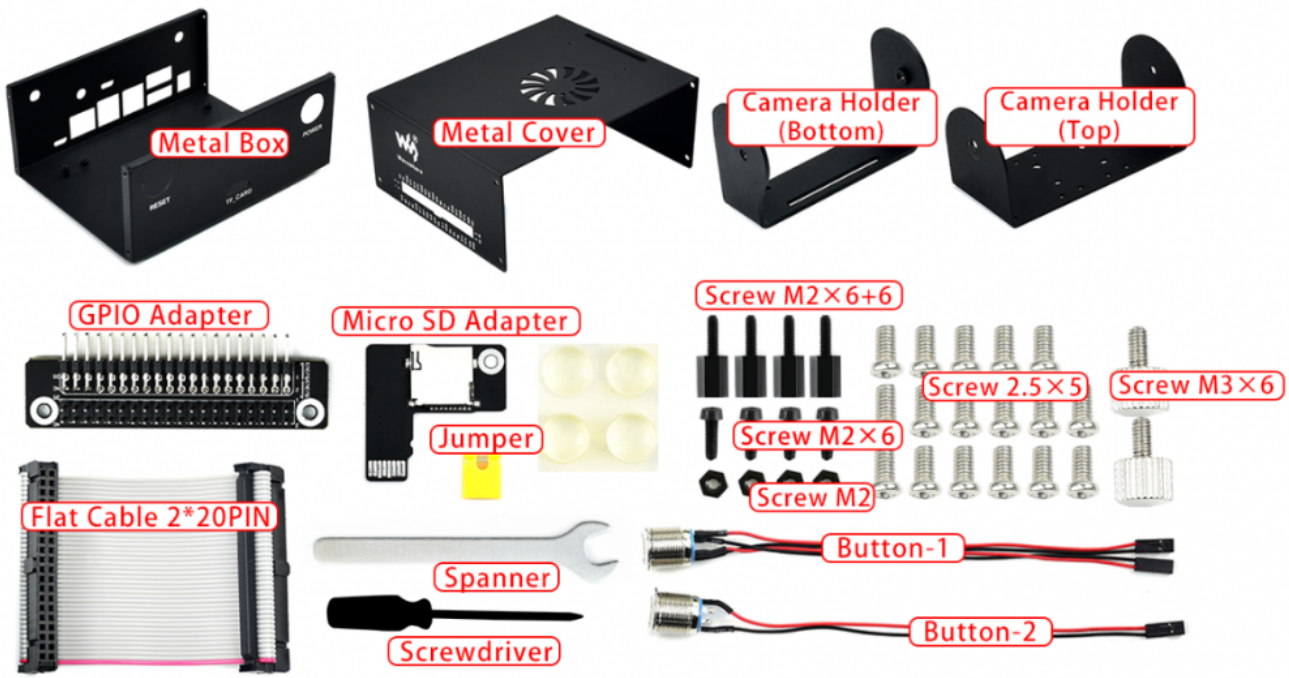


Abbildung: Waveshare Case

NVIDIA Jetson Nano Developer Kit A02 model I

NVIDIA Jetson Nano has 2 models in J1010 there are 2 Type C ports and in standard NVIDIA Jetson Nano consists of 1 Micro USB and 1 Type C port.NVIDIA, 2020b

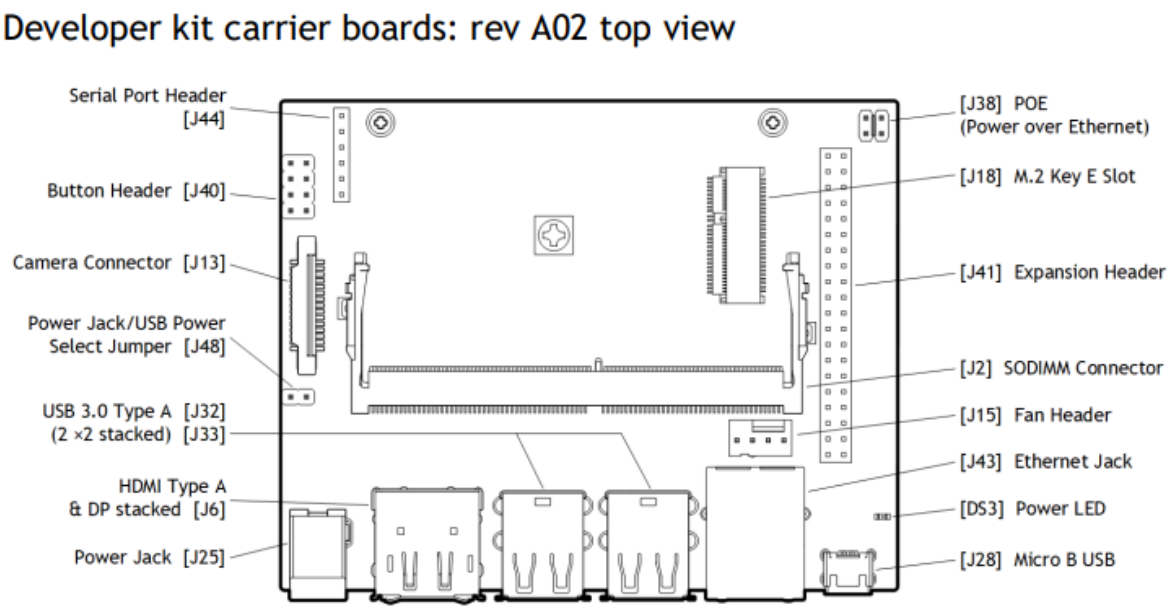


Abbildung: NVIDIA Jetson Nano Developer Kit 2GB (original A02 model)

Power and Reset Connections I

[J40] **Carrier board rev A02 only**: 8-pin button header; brings out several system power, reset, and force recovery related signals (see the following diagram).

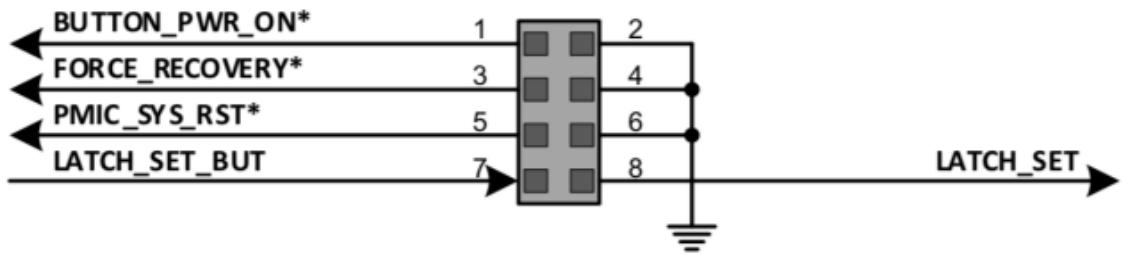


Abbildung: [J40] Header Pins of NVIDIA Jetson Nano Developer Kit A02 model

- Pins 7 and 8 disable auto power-on.
- Pins 1 and 2 initiate power-on if auto power-on is disabled.
- Pins 5 and 6 reset the system.

Power and Reset Connections II

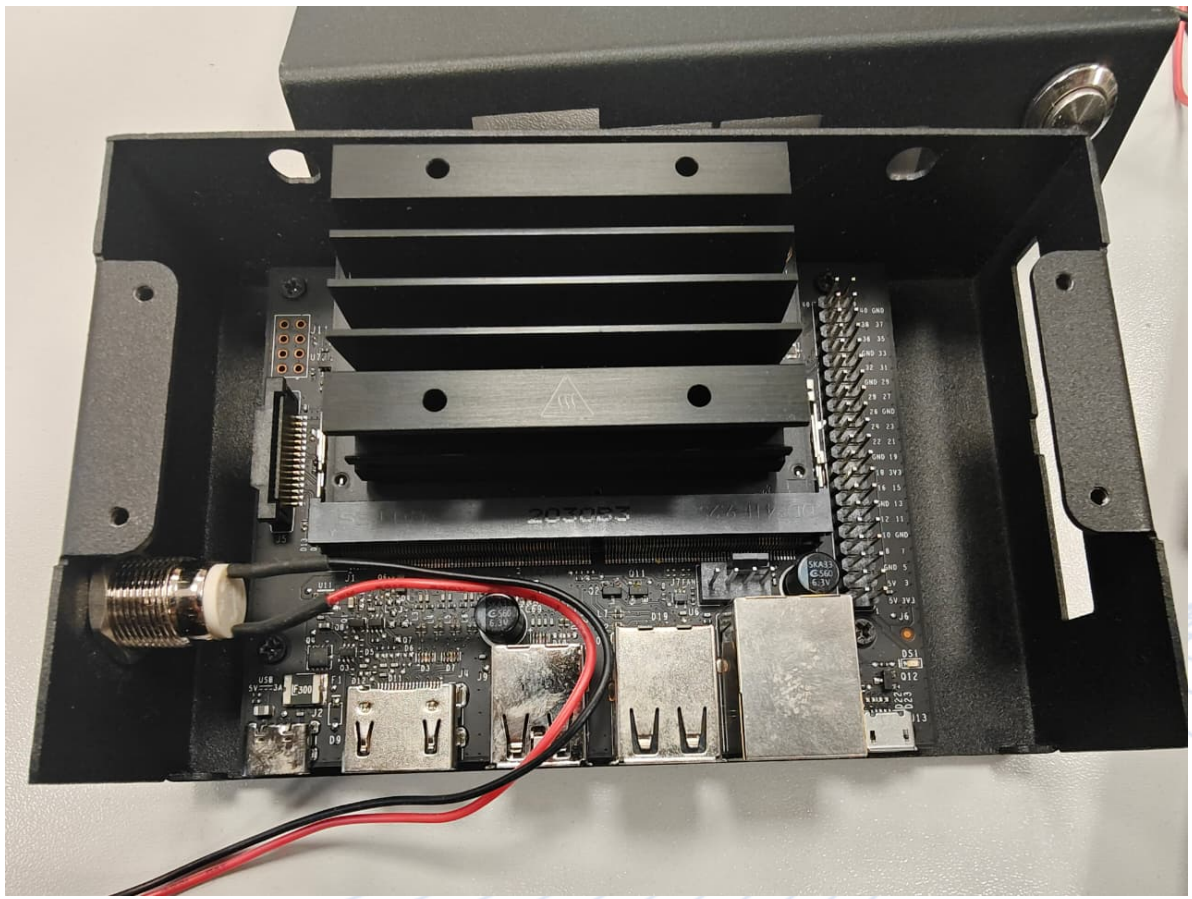


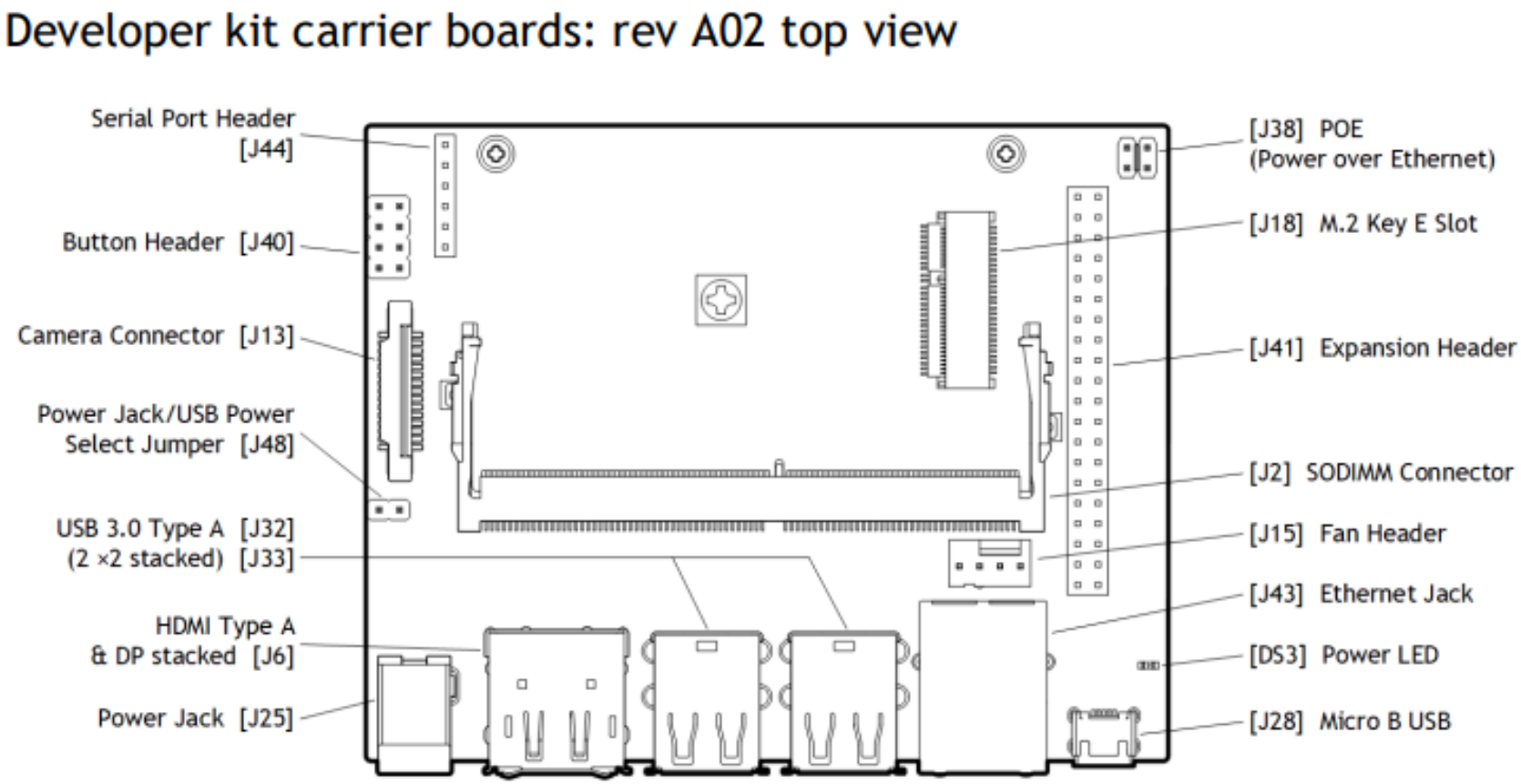
Abbildung: NVIDIA Jetson Nano Developer Kit A02 model inside Waveshare Case

Power and Reset Connections III

[J41] 40-pin expansion header includes:

- Power pins. Two 3.3V power pins and two 5V power pins. These are not switchable; power is always available when the developer kit is connected to power. Two 5V pins can be used to power the developer kit at 2.5A each.
- Interface signal pins, All signals use 3.3V levels.
- By default, all interface signal pins are configured as GPIOs, except pins 3 and 5 and pins 27 and 28, which are I^2C SDA and SCL, and pins 8 and 10, which are UART TX and RX. L4T includes a Python library, `Jetson.GPIO`, for controlling GPIOs. NVIDIA, 2020b

NVIDIA Jetson Nano Developer Kit (Rev A02)— Interface I



NVIDIA Jetson Nano Developer Kit (Rev A02)— Interface II

J501	Slot for a microSD card.
Heatsink	Passive heatsink supporting 10W module power at 25°C ambient.
DS3	Power LED indicating system power-on.
J2	SO-DIMM connector for Jetson Nano module.
J6	HDMI and DisplayPort connector stack.
J13	Camera connector (CSI), supports IMX219 modules.
J15	4-pin fan control header with PWM and tachometer support.
J18	M.2 Key E slot with PCIe x1, USB 2.0, UART, I2S, and I2C.
J25	5V 4A barrel jack power input (2.1×5.5×9.5mm plug).
J28	Micro-USB 2.0 for 5V 2A power input or Device Mode.
J32	USB 3.0 Type A (two stacked).
J33	USB 3.0 Type A (two stacked).

NVIDIA Jetson Nano Developer Kit (Rev A02)— Interface III

J38 Power over Ethernet header exposing DC voltage from RJ45.

J40 8-pin button header for power, reset, force recovery.

J41 40-pin expansion header with 3.3V signals and power pins.

J43 RJ45 Gigabit Ethernet connector.

J44 3.3V serial port header (UART console).

J48 Power select jumper for choosing J28 or J25 as power source.

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Next Steps

Next Steps on NVIDIA Jetson Nano Developer Kit I

- Note the tests that are successful
- Check for version of Jetpack and document it.
- Calibrate the External Sensors and components based on the requirements of project.
- Camera used in this project
- JetBot is an open-source AI project for makers, students and enthusiasts who are interested in learning AI and building fun applications. JetBot is a great launchpad for creating entirely new AI projects. NVIDIA, 2020b

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Vielen Dank

für Ihre Aufmerksamkeit

Quellen

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